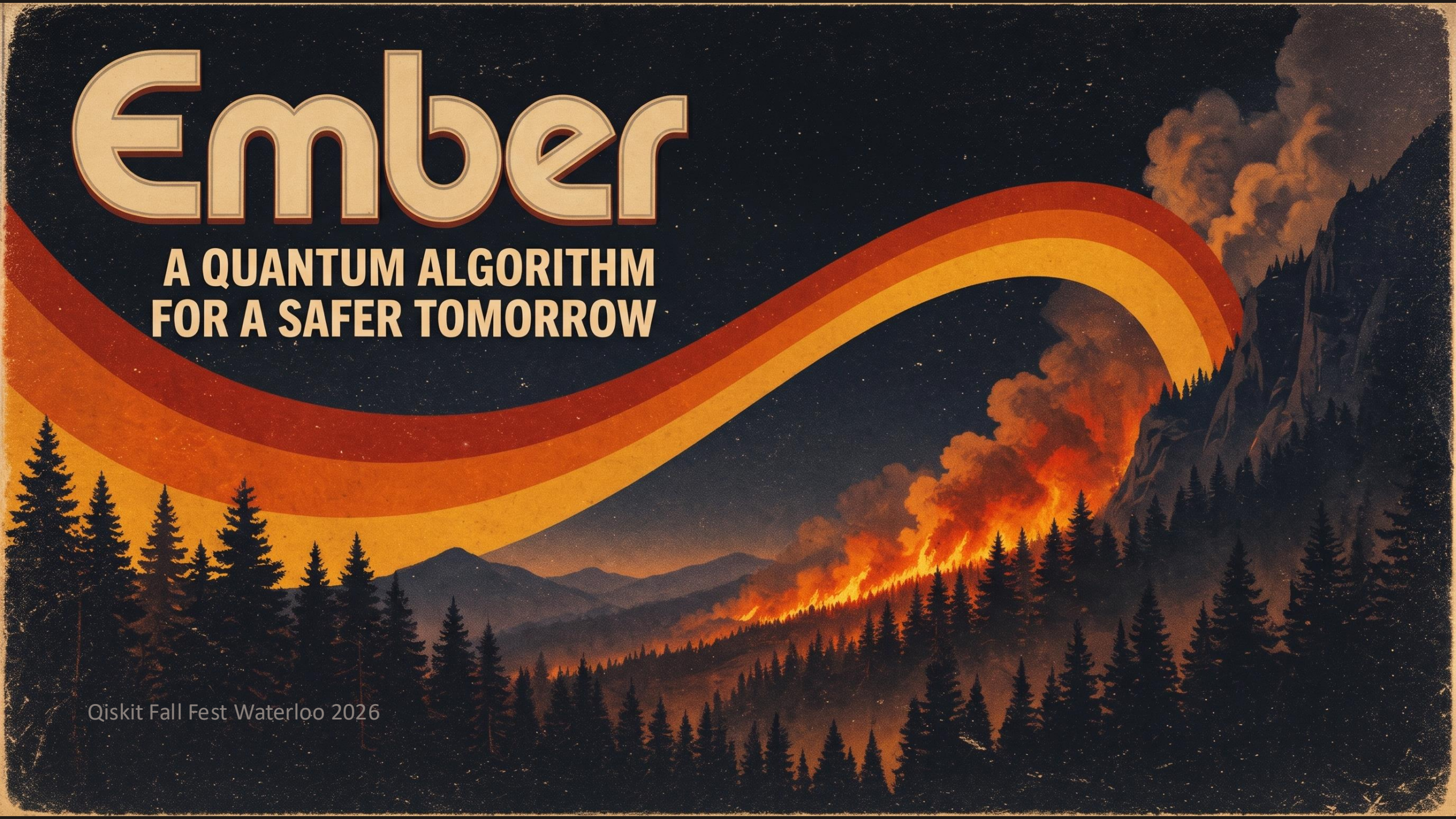

Ember
A QUANTUM ALGORITHM
FOR A SAFER TOMORROW
Qiskit Fall Fest Waterloo 2026

2023 was Canada's worst fire season

17.2 M ha

burned in Canada in 2023, a national record

Source: CIFFC, Canada Report 2023

~20,000

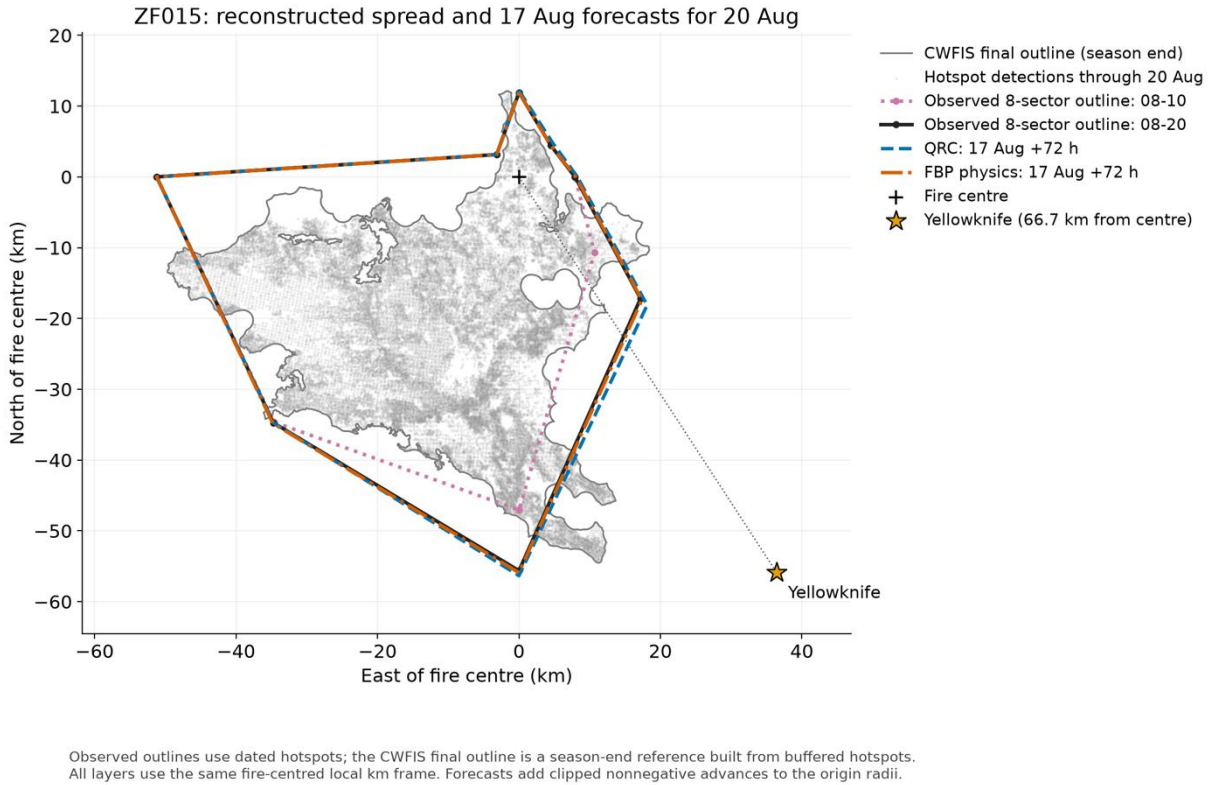
people ordered out of Yellowknife on 16 Aug 2023

Source: CBS News, 16 Aug 2023

Where will the fire edge be in 24 h and in 72 h?

Ember forecasts the edge advance in 8 directions, starting on day 3 of a fire.

We rebuild each fire's daily outline from satellites



Yellowknife fire ZF015: rebuilt outlines and 17 Aug forecasts for 20 Aug

1

Hotspots to fires

CWFIS satellite detections, grouped by each fire's final perimeter

2

Daily outline in 8 sectors

Edge distance from the day-1 centre; growth through 2 km chains, at most 30 km a day

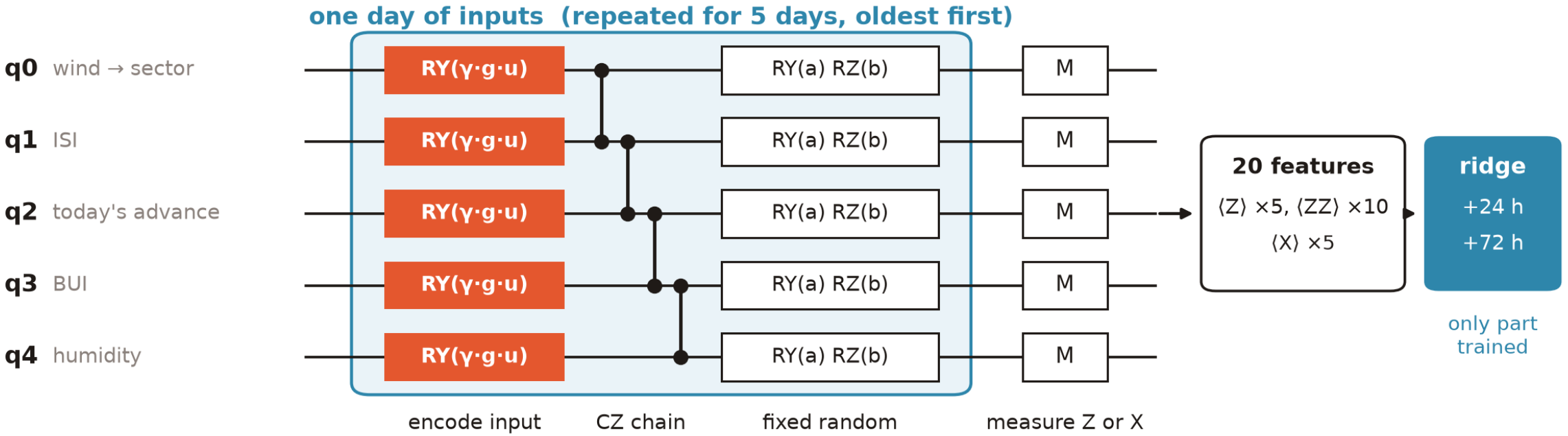
3

Forecast

Edge advance in each sector at +24 h and +72 h, from day 3

Train 2018–21: 378 fires
Choose settings 2022: 102 · Test 2023: 392, scored once

First model: 5 qubits



One qubit per input

Wind toward the sector, ISI, today's advance, BUI, humidity

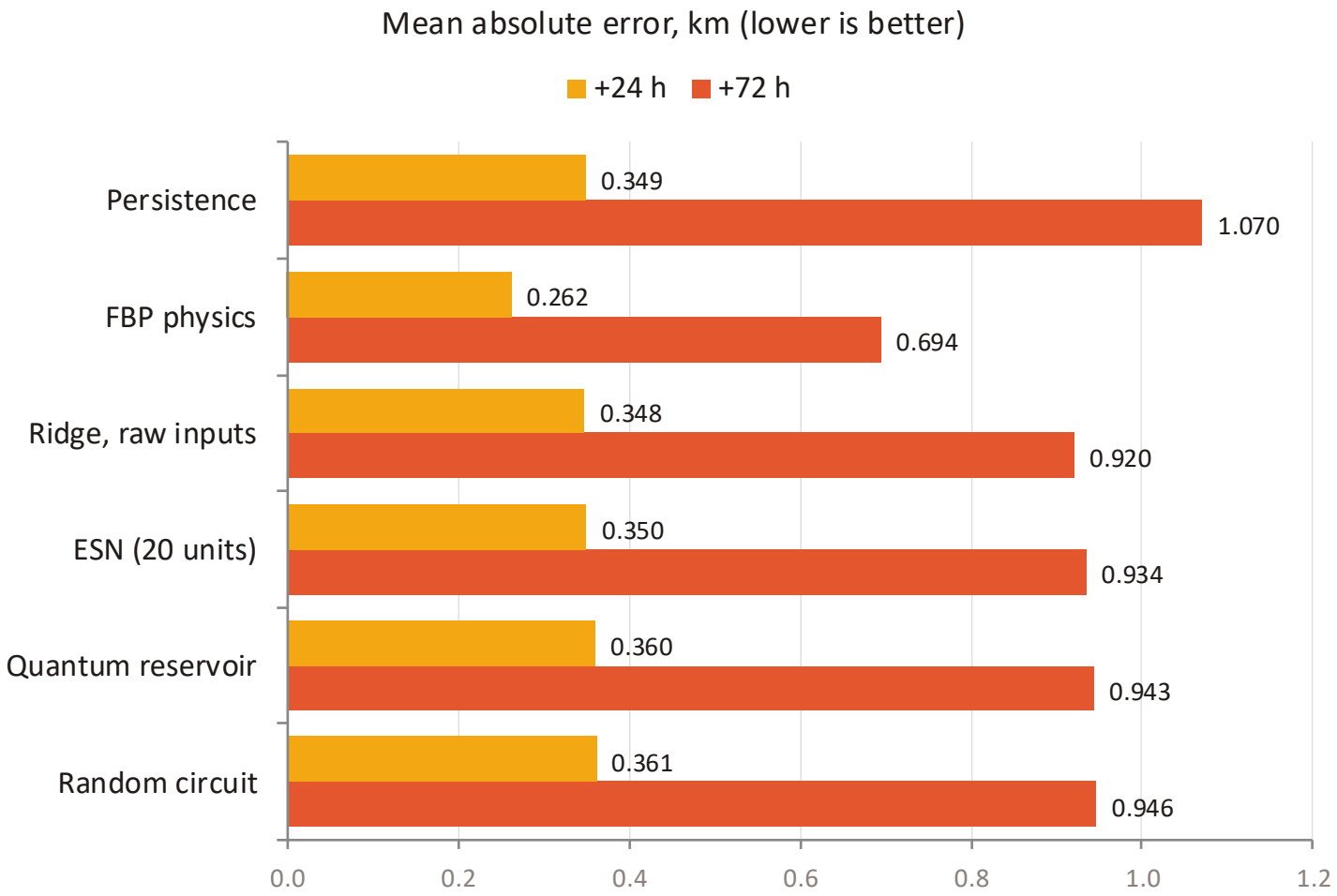
Fixed circuit

20 CZ gates (4 gates x 5 days). Nothing quantum is trained; only the ridge read-out

Chosen in 2022 only

5 days, $\gamma = \pi/4$, ridge $\lambda = 10^{-6}$, locked before testing

Test 2023: quantum ties classical models



392 test fires, scored once. Values at +24 h / +72 h. No quantum advantage is claimed.

About level with classical

The 5-qubit quantum reservoir performs almost identically to classical machine learning models (Ridge, ESN) and completely random circuits.

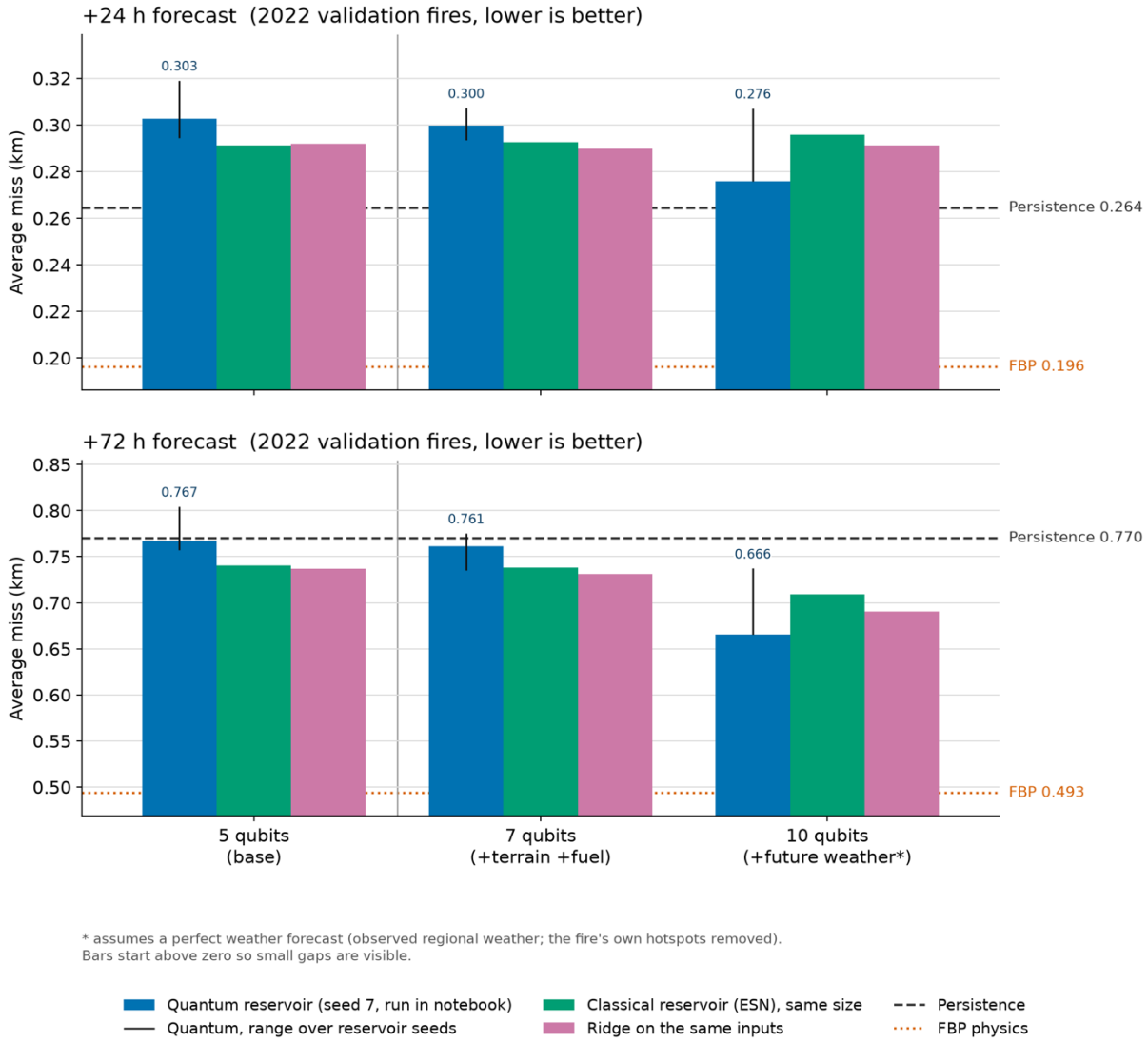
FBP wins on mean error

Canada's official model beats all ML (0.262 / 0.694 km). It uses the physical fuel type and future weather forecasts—hinting at what our model might be missing.

72h is the real test

At 24h, guessing the fire will move at the exact same speed as yesterday (Persistence) is highly accurate. The quantum and classical ML models only beat this simple baseline at the 72h mark.

Second model: upgrading the quantum resevoir



Fuel & Terrain show little impact

Expanding from 5 to 7 qubits to include terrain and fuel type barely moved the needle. The quantum model continued to lag slightly behind the classical machine learning baselines.

Future weather flips the script

Expanding to 10 qubits to include future weather forecasts finally changed the dynamic. With this data, the quantum reservoir began to slightly outperform the classical reservoir and Ridge.

FBP's physical complexity still wins

Even at 10 qubits, FBP remains the best. Unlike ML models that just match data patterns, FBP uses highly complex, empirical non-linear equations.

On ibm_fez, the average error survive the noise

238 s

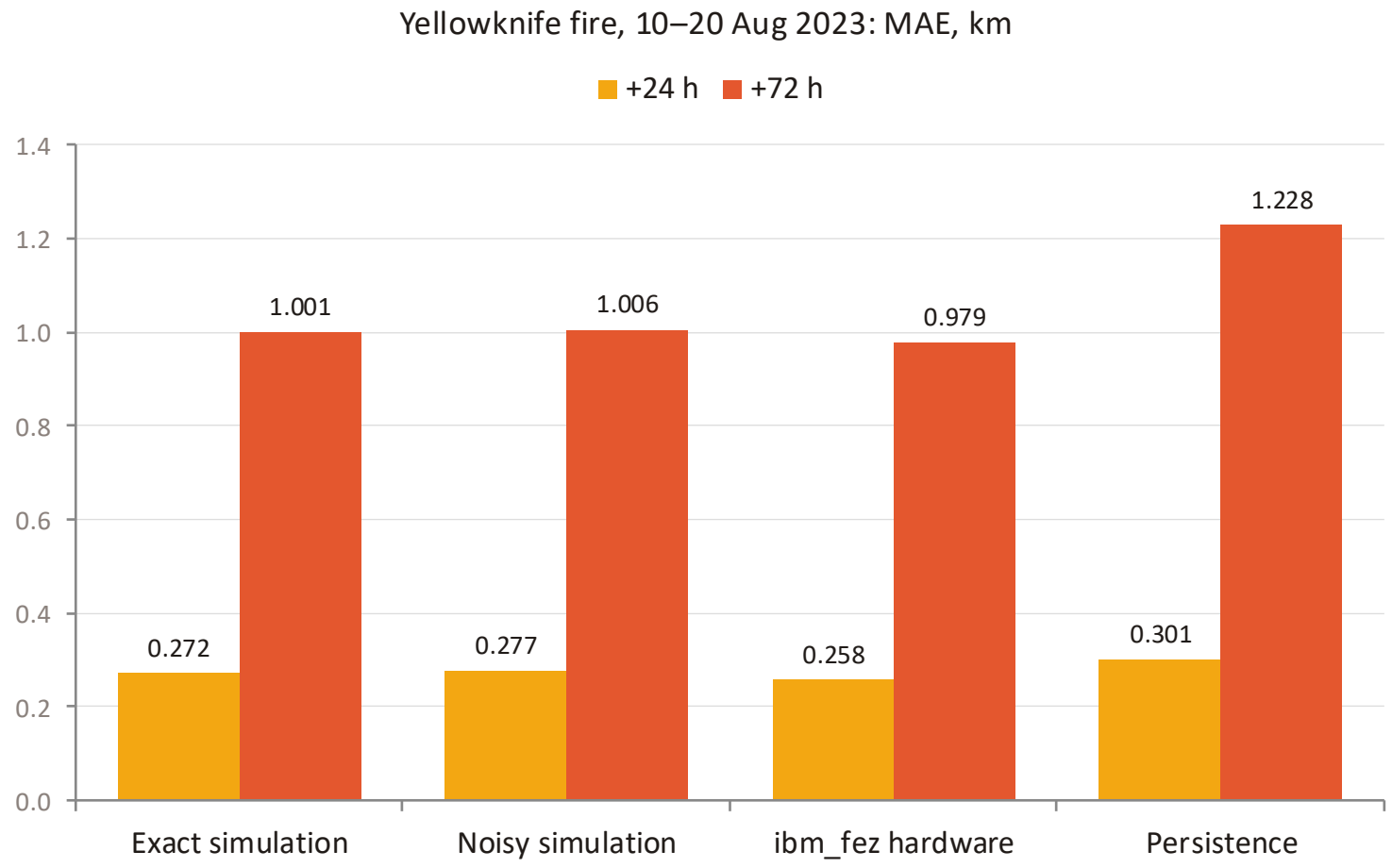
QPU time for all 88 circuits

10,000

shots per circuit

20 CZ

per circuit



88 correlated sector windows from one fire; descriptive only. The same frozen read-out is applied to every source.

Takeaways

What we showed

- Daily fire outlines rebuilt from satellite hotspots
- An honest test: quantum reservoir generally ties classical ones
- Weather forecasts could give quantum a slight edge over classical ML
- Forecasts hold up on real IBM hardware

Limits

- Only one fire run on hardware
- The 30 km/day cap truncates extreme blow-ups
- FBP empirical physics equations still win
- Idealised weather forecasts were used
- Cloud cover and smoke not considered

Next

- Hybridize with FBP: merge quantum pattern learning with FBP physics
- Run more fires on real hardware
- Rank nearby towns by forecast closest approach

Quantum reservoirs match classical ML today; hybridizing with real physics is the path to tomorrow.